

The Relationship between Banking Sector Development and Innovation Activities in Countries in the European Union from 2010 to 2020

Diploma Thesis

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1. INTRODUCTION

In recent years, fierce competition exists in almost industries and its entities are enforced to continuously improve itself to adapt to non-stop changes. Therefore, innovation activities are considered as a vital element leading to success in every field of recent society.

Innovation is not just a trendy buzzword that is casually thrown around in business meetings. It is a tangible, dynamic process that involves transforming a novel idea or invention into a valuable product or service. This product or service then becomes something for which customers are willing to pay. Innovation is the deliberate application of information, imagination, and initiative aimed at extracting greater or different values from resources. In this context, it plays a pivotal role in offering distinctive solutions to the vast array of challenges that businesses encounter daily.

However, the path to successful innovation is not without its hurdles. It mandates the cultivation of a culture that encourages creativity and risk-taking, which can be an arduous task to implement within an organization. The fostering of such a culture calls for an environment where employees feel safe to share and explore new ideas. Moreover, the innovation process often demands substantial commitments of time and resources, elements that some businesses may find discouraging. These significant investments can sometimes act as a deterrent, especially for smaller businesses or start-ups that are already working within tight budgets.

Despite these challenges, the potential benefits that can be reaped from innovation significantly outweigh the obstacles. Businesses that welcome and embrace innovation can gain a competitive edge through enhanced operational efficiencies, the establishment of new markets, and the delivery of superior value to customers. These benefits can drastically improve a business's market position and overall profitability.

In today's business climate, characterized by relentless competition and constant change, innovation is not just a desirable strategy but an absolute necessity. Businesses cannot afford to be complacent or to rest on their laurels. They must continually strive to improve, innovate and evolve in response to shifting market dynamics. If they fail to do so, they run the risk of stagnation and ultimately, obsolescence. Therefore, in the face of such a competitive environment, businesses must view innovation as a crucial ingredient in their recipe for success.

Apparently, European countries are in the above tendency as well. A number of researches related to innovation activities was conducted. Kitukutha et al. (2021) provided that the Covid-19 pandemic creates an increase in demand for digital financial products. In other words, this period must be innovated immediately to meet changing needs because of the pandemic. Lozano-Vivas and Pasiouras (2014) found that financial innovation by boosting off-balance-sheet activities in banking system has a positive impact on profit efficiency. Conversely, the development in using innovative technologies of financial institutions and fintech companies results in an increase of cyber attacks (Petrushenko, Yuriy et al., 2018; Syniavska, Olga et al., 2019). Other current studies examined other determinants which have a relationship with innovation activities in general such as “competitiveness of manufacturing industry in developing markets; income redistribution; environmental, social and governance practices; etc.” (Padilla-Lozano et al., 2021; Brzezinski and Michal, 2022; Dicuonzo, Grazia et al., 2022).

As can be seen from above researches, innovation is indeed an indispensable element in every area. Bank system is one of the most noticeable sectors where definitions such as “repetition, similarity, traditionality” need to be limited. This could be explained by rapid alterations in customer’s need. For instance, nowadays customers prefer online payment to cash, firms are more confident to globalize its products thanks to new created financing methods from banking system, etc. Innovation activities, hence, are undoubtedly essential for banking system’s growth.

In today's fast-paced and interconnected world, the banking system serves as the backbone of our economic structure. As such, it plays an essential role in our daily lives, whether in personal finance management, business transactions, or even governmental fiscal operations. Given this ubiquitous nature and the consequential impact on a vast array of stakeholders, it becomes crucial to delve deeper into understanding the propelling forces that necessitate a relentless pursuit of innovation within the banking sector. These catalysts for change can be broadly distinguished into two main categories: technological advancements and evolving customer expectations.

The dawn of the technological era, characterized more specifically by the widespread adoption of the Internet and mobile devices, has brought about a dramatic transformation in the banking landscape. As technology permeates every aspect of our lives, it has unfolded new paradigms in customer behaviors and expectations. Today's tech-savvy customers, who enjoy the convenience and immediacy offered by digital platforms across various spheres of their lives, now anticipate similar accessibility when it comes to banking services. They expect to be able to access their financial information and perform banking transactions anytime, anywhere, and from any device.

This profound shift in consumer behavior and expectations has created an exigent need for the banking industry to develop and implement innovative solutions, which are aligned with the digital age. Such solutions encompass online banking platforms, mobile banking applications, and digital wallets, all designed to provide customers with the seamless, convenient experience they demand. Banks that lag in adapting to these technological advancements, or fail to invest in innovation, run the risk of losing their customer base to competitors that are more technologically progressive and responsive to the digital wave.

Yet, technological advancements represent just one side of the coin. The other side is dominated by the significant evolution of customer expectations. Modern customers, particularly the younger generation that has grown in the cradle of the digital age, are no longer satisfied with just basic banking services. They aspire for more. They expect personalized services that cater not just to their generic needs but are tailored to their individual preferences and financial goals.

This heightened customer expectation has given rise to a new breed of innovative banking services. These include robo-advisors, which provide automated, algorithm-driven financial planning services with little to no human supervision, and AI-based customer service bots, designed to offer prompt and personalized customer service. Such technologies are not just revolutionizing the way banking services are delivered, but they are also redefining the very essence of customer experience in the banking sector. Banks that can skillfully harness the power of these technologies to deliver superior, personalized customer experiences stand a higher chance of gaining a competitive edge in the cutthroat banking industry.

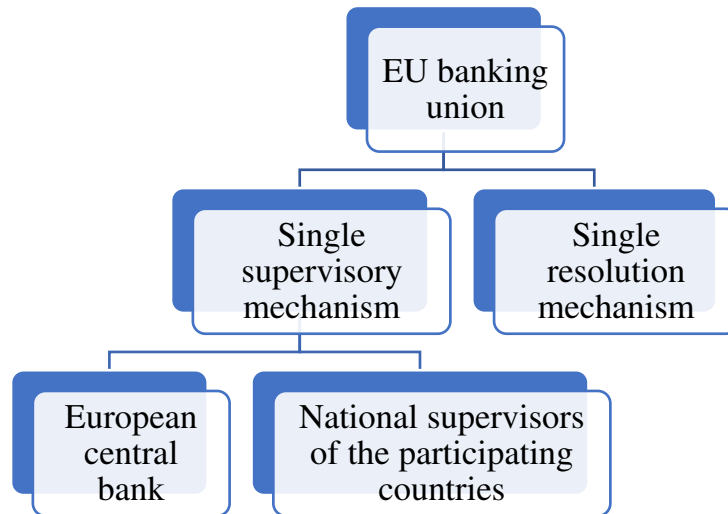
To summarize, the continuous need for innovation in the banking sector is driven by the symbiotic interplay of technological advancements and changing customer expectations. As these forces continue to evolve, banks must remain agile, embracing change and innovation, to retain their relevance and competitiveness in the ever-evolving banking landscape.

In reverse direction of perspective, the development of financial institutions in general and banking sector in specific is also expected to have a positive impact on innovation. Empirically, Clò, Stefano et al. (2022) represented that patent application as a proxy for innovation increases thanks to financial support from European development financial institutions. Similarly, Tee, Lain-Tze et al. (2014) indicated a crucial role of banking system development in boosting innovative activities in selected east Asian countries.

Based on mentioned studies above, this paper is devoted to the relationship between the development of banking sector and innovation activities in European Union (EU) countries in the period of time from 2010 to 2020.

However firstly, it is necessary to have background knowledge of EU banking sector³. EU banking union was created with some main purposes. First, it creates the environment for its participants to be under common rules and administrative standards. Second, treating equally to national and cross-border banking activities. Third, timely detect and prevent participants as banks from failing. According to EU banking union's structure, it can be displayed as the diagram below:

Diagram 1. EU banking union's structure



Source: Own processing

Single supervisory mechanism has some missions such as ensure safety of the European banking system, enforce financial integration and stability, guarantee consistent supervision. Single resolution mechanism, while, is to maintain effectively resolution of failing banks.

³ Bank, E. C. (2022, September 22). *ECB Banking Supervision - SSM*. European Central Bank - Banking Supervision. Retrieved September 29, 2022, from <https://www.bankingsupervision.europa.eu/home/html/index.en.html>

Besides EU banking system, this chapter introduces the measurement of innovation activities. The number of patent applications was historically considered as a reasonable reflection on enhancement, improvement and development of almost aspect of society. It is a legal recognition for new discovery which has not been detected before. Therefore, a number of studies from the past until recent years have used the number of patent applications as a proxy for the activity of innovation such as Ang, J. B. (2010), Tee, Lain-Tze et al. (2014), Burhan et al. (2017), Clò, Stefano et al. (2022). Similarity, this paper applies measurement for innovation activities by patent applications volume.

The remaining parts are organized as the following. Chapter 2 provides the objectives of this paper. Chapter 3 reports collected sample data and methodology. Chapter 4 presents literature review and hypothesis development. Next, findings are revealed and discussed in chapter 5. The paper finishes with conclusions and proposals in chapter 6.

2. OBJECTIVE

This research primarily aims to explore and identify the potential positive impacts that the evolution and development of the banking sector might have on innovation activities within the countries of the European Union. To accomplish this, the research will employ a data-driven approach, leveraging information collected over a significant time period, from 2010 to 2020. This approach is planned in order to ensure a comprehensive and accurate analysis of the trends and patterns associated with the development of the banking sector and its effects on innovation activities in these countries.

Furthermore, this research paper seeks to add to the existing body of literature by providing insights into the influence of the growth and development of financial markets on domestic innovation activities inside these countries. This aspect of the research will contribute to the broader understanding of the interplay between financial markets and innovation activities, thereby enriching the understanding within this field of study.

In order to efficiently navigate this research, several key research questions are formulated. These questions serve as guiding posts and enable the research to adequately address the research objectives:

- What are the most accurate and reliable indicators that can effectively represent the progression and development of a banking system and innovation activities within a country in the European Union? This question aims to identify the metrics and indicators that best represent the banking and innovation landscape within these countries.
- How does the banking system of a country within the European Union influence its activities associated with innovation? This question seeks to understand the mechanisms through which a country's banking system can impact innovation-related activities.

- What is the nature and extent of the relationship between a country's banking market within the European Union and its broader financial markets? This question aims to dissect the relationship between the banking market and the wider financial market within an EU country. The goal is to understand the dynamics of this relationship and how it affects innovation activities.

These carefully crafted research questions are designed to ensure a comprehensive and thorough exploration of the research topic, thereby facilitating an in-depth understanding of the subject matter.

3. METHODOLOGY

This chapter provides description of sample data in details and its processing. Besides, the research model and research method are also introduced to deal with collected data.

3.1. Sample data selection and collection

The research uses a panel data of 24 out of 27 EU countries⁴ collected from the World bank open data⁵ in the period from 2010 to 2020, where 03 EU countries which are Cyprus, Ireland and Slovenia are dropped out due to huge amount of data shortage. Besides, some missing values are dealt by average method and based on the continuity of time series data to assure the sample of research panel data to be strongly balanced.

According to variables associated with the development of banking sector, they are applied a 1-year lag to observe more exactly its effects on the number of patent applications. Therefore, its data is collected from 2009 to 2019.

Besides, most of research variables in this paper are measured by the natural logarithm to decrease the volatility in figures. “Research and development expenditure” and “The governance quality of a country” are the two variables which are not adjusted in logarithmic form due to their minor volatility in values.

⁴ Country profiles. European Union. (n.d.). Retrieved December 29, 2022, from https://european-union.europa.eu/principles-countries-history/country-profiles_en

⁵ Atamanov, A., Blankespoor, B., Nagle, P., & Baffes, J. (2022, December 21). World Bank Open Data. Data. Retrieved December 30, 2022, from <https://data.worldbank.org/>

3.2. Variables description and measurement

Innovation activities measurement

As stated in the “Introduction” part, “the number of patent applications” is a reliable indicator representing for innovation activities. Hence, this research employs it in logarithmic form (**lpa**) as the dependent variable.

The World Intellectual Property Organization (WIPO) provided a background knowledge about patent applications as a recognition for exclusive rights for a creative invention and a protection for the patent owner in generally 20 years.

A patent is a legal recognition granted to an inventor or assignee, conferring exclusive rights to the invention for a specified period of time. This recognition serves as an incentive for innovation and creativity, as it offers protection against unauthorized use, reproduction, or commercial exploitation of the invention.

The patent system operates on the fundamental principle of disclosure. In exchange for the exclusive rights granted to the patent holder, the inventor is required to disclose the details of the invention in a patent application. This disclosure serves a dual purpose: it promotes the dissemination of knowledge and technical information, while also facilitating the examination and evaluation of the invention by patent offices.

A patent application typically includes a detailed description of the invention, accompanied by relevant drawings, claims, and any other necessary documentation. The description should be sufficiently clear and complete to enable a person skilled in the relevant field to carry out the invention. It should also disclose the best mode known to the inventor for carrying out the invention.

Upon submission of a patent application, the process of examination begins. Patent offices, such as the United States Patent and Trademark Office (USPTO) and the European Patent Office (EPO), review the application to assess its compliance with legal requirements and its patentability. This includes evaluating the novelty, inventive step, and industrial applicability of the invention.

The examination process involves a rigorous review of prior art, which refers to all publicly available information that existed before the filing date of the patent application. Prior art includes existing patents, scientific publications, and other forms of publicly disclosed information. The purpose of this review is to determine whether the invention meets the criteria of novelty, non-obviousness, and industrial applicability.

In addition to the examination of patent applications, patent offices also play a crucial role in the dissemination of patent information. Patent databases, maintained by these offices, provide a comprehensive repository of technical knowledge and innovation. These databases enable inventors, researchers, and businesses to access valuable information, conduct prior art searches, and assess the state of the art in a particular field.

Furthermore, patent offices collaborate with international organizations, such as WIPO, to promote harmonization and standardization of patent laws and procedures. This collaboration ensures that inventors and innovators receive consistent and reliable protection for their intellectual property rights, regardless of their geographic location.

In conclusion, patent applications serve as a vital mechanism for the recognition and protection of creative inventions. Through the disclosure of technical knowledge and the granting of exclusive rights, patents foster innovation, encourage research and development, and contribute to economic growth. The efforts of organizations like WIPO and patent offices around the world are instrumental in maintaining a fair and effective patent system that benefits inventors, businesses, and society as a whole.

Banking sector development measurement

Following Beck et al. (2010), two potential indicators evaluating the development of banking system are “Bank credit to bank deposits” and “Deposit money banks assets to GDP⁶”. Specifically, “Bank credit to bank deposits” is as a share of total deposits provided to the private sector by deposit money banks which include commercial banks and other financial institutions that accept transferable deposits. “Deposit money banks assets to GDP”, while, reveals total assets held by deposit money banks as a share of GDP.

Correspondingly, Tee, Lain-Tze et al. (2014) applied the same measurement method as Beck et al. (2010) to evaluate banking sector development, but with two more indicators which are “Private credit by deposit money banks to GDP” and “Private credit by deposit money banks and other financial institutions to GDP”.

Based on the above studies, this research measures banking sector development expressed in logarithmic form (**lbsd**) via four indicators: Bank credit to bank deposits (**ldeposit**), Deposit money banks assets to GDP (**lasset**), Private credit by deposit money banks to GDP (**lbank**), Private credit by deposit money banks and other financial institutions to GDP (**lcredit**).

The indicator "Bank credit to bank deposits" is an important measure that reflects the lending activities of banks in relation to the deposits they hold. It indicates the extent to which banks are utilizing their deposits to provide loans and credit to individuals, businesses, and other entities. A higher ratio suggests that banks are actively extending credit, which can stimulate economic growth and investment. On the other hand, a lower ratio may indicate a more conservative lending approach by banks, which could have implications for economic expansion.

⁶ Gross domestic product

Another key indicator, "Deposit money banks assets to GDP", provides an assessment of the size and significance of the banking sector in relation to the overall economy. It measures the total assets held by deposit money banks, including loans, investments, and other financial instruments, as a percentage of the country's gross domestic product (GDP). A higher ratio suggests a larger banking sector relative to the size of the economy, indicating a significant role played by banks in providing financial intermediation services and facilitating economic activities.

The indicator "Private credit by deposit money banks to GDP" focuses specifically on the credit extended by deposit money banks to the private sector. It measures the amount of credit provided by banks to individuals and businesses, relative to the size of the economy. This indicator helps assess the availability and accessibility of credit for private sector activities, which are crucial for investment, entrepreneurship, and economic development. A higher ratio indicates a greater level of credit availability, which can support economic expansion and productivity growth.

Lastly, the indicator "Private credit by deposit money banks and other financial institutions to GDP" provides a broader view of the overall credit market by including credit extended not only by deposit money banks but also by other financial institutions. This indicator captures the combined lending activities of banks and non-bank financial institutions, giving a more comprehensive measure of credit provision in the economy. It reflects the overall depth and efficiency of the credit market, which is essential for supporting economic activities and financial stability.

These indicators collectively offer valuable insights into the development and performance of the banking sector in a country. By analyzing these indicators over time, policymakers, regulators, and stakeholders can assess the strengths and weaknesses of the banking system. This information can inform policy decisions, regulatory reforms, and strategic initiatives aimed at fostering a sound and resilient banking sector that contributes to sustainable economic growth.

In conclusion, the development of the banking sector in a country can be assessed through various indicators, including "Bank credit to bank deposits", "Deposit money banks assets to GDP", "Private credit by deposit money banks to GDP", and "Private credit by deposit money banks and other financial institutions to GDP". These indicators provide insights into the lending activities of banks, the size and significance of the banking sector, the availability of credit for the private sector, and the overall depth of the credit market. By monitoring and analyzing these indicators, policymakers and stakeholders can make informed decisions to promote a robust and stable banking sector that supports economic growth and financial stability.

Control variables measurement

Besides the main variables associated with research elements which are innovation activities and banking sector development, this paper complements control variables which have potentially positive effects on innovative activities. The purpose is to limit the influence of confounding and establish the true relationships between the mainly researched independent and dependent variables.

The first control variable is “GDP per capita” displayed in current international dollars applying purchasing power parity (PPP), expressed in logarithmic form (**lgdp**). This figure determines a country’s wealth which is anticipated to enhance innovative activities in quality (Teitel, 1994).

GDP per capita, when measured in current international dollars using purchasing power parity (PPP), serves as a critical indicator of a country's economic wealth. This metric not only reflects the average income and living standards of a nation's population but also provides insights into its potential for fostering innovative activities and promoting overall quality of life.

By taking into account the purchasing power of currencies across different countries, GDP per capita based on PPP offers a more accurate representation of a nation's economic strength and the real value of its output. It helps to level the playing field and allows for meaningful comparisons between countries, regardless of exchange rate fluctuations or disparities in cost of living.

A higher GDP per capita signifies a greater capacity for investment in research and development, education, infrastructure, and other factors that contribute to innovation. It provides a solid foundation for nurturing a conducive environment that fosters creativity, attracts talent, and supports the growth of industries and businesses. With increased wealth, countries can allocate resources towards the development of cutting-edge technologies, scientific advancements, and the creation of sustainable solutions to societal challenges.

Moreover, a high GDP per capita indicates a higher overall standard of living, as it reflects the ability of individuals to afford goods and services in the domestic market. This, in turn, can lead to a higher demand for innovative products and services, spurring further investment in research and development. Additionally, a wealthier population often enjoys better access to education, healthcare, and other essential services, which can contribute to the creation of a highly skilled workforce and a more productive society.

Next is the second control variable, “Research and development (R&D) expenditure expressed as a percent of GDP” (**rd**), is the gross domestic expenditures on R&D compared to GDP. This indicator attaches to basic research, applied research and experimental development which create more effective applications based on new detected theories. For instance, national R&D program in Korea is considered as a long run strategy for developing fuel cell electric vehicle industry which then contributing to the national innovation system (Yoo and Park, 2022).

Apparently, "Research and development (R&D) expenditure expressed as a percent of GDP" is a widely used metric to measure the level of investment in research and development activities within a country. It provides valuable insights into the commitment and prioritization of R&D within the overall economy.

By expressing R&D expenditure as a percentage of GDP, we can understand the relative scale of investment in R&D compared to the size of the economy. This metric allows for cross-country comparisons, as it takes into account the differences in the size and structure of economies. It serves as an indicator of the level of resources allocated to R&D and reflects the importance given to innovation and technological advancement.

Monitoring the R&D expenditure as a percent of GDP over time can reveal trends and patterns in research and development investments. An increase in this ratio may indicate a growing emphasis on innovation and a commitment to staying competitive in a globalized world. Conversely, a decline in the ratio may signal a lack of investment in R&D or a shift in priorities.

Governments, policymakers, and organizations often use this metric to assess the effectiveness of their R&D policies and strategies. It helps them evaluate the impact of R&D investments on economic growth, productivity, and competitiveness. By benchmarking against other countries or regions, they can identify areas for improvement and make informed decisions regarding resource allocation.

Furthermore, the R&D expenditure as a percent of GDP can provide insights into the overall innovation ecosystem of a country. It reflects the collaboration between academia, industry, and government in fostering research and development activities. A higher ratio suggests a strong and supportive innovation ecosystem that encourages knowledge transfer and the commercialization of research outcomes.

The third control variable is “the gross ratio school enrollment in tertiary education”, utilized as a proxy for a country’s knowledge, expressed in log-transformed (**ledu**). This variable is calculated by the ratio of total enrollment to the population of the official age group corresponding to the education of tertiary. This research applies “ledu” as a control variable based on a recent study of Moon et al. (2019), indicating that expected higher performance of innovation outcomes corresponding to the higher education level.

The significance of the gross ratio of school enrollment in tertiary education lies in its strong correlation with innovation outcomes. It is widely recognized that countries with higher levels of tertiary education tend to achieve higher performance in terms of innovation. This is because a well-educated population possesses the necessary skills, knowledge, and critical thinking abilities to drive innovation and contribute to technological advancements.

When a country places a strong emphasis on tertiary education and achieves a high gross ratio of school enrollment in this domain, it signifies a commitment to nurturing a knowledgeable society. Individuals who pursue tertiary education are equipped with advanced subject-specific knowledge and are exposed to a diverse range of ideas and perspectives. This exposure fosters creativity, encourages critical thinking, and cultivates the ability to generate innovative solutions to complex problems.

Moreover, a high gross ratio of school enrollment in tertiary education indicates that a country values the importance of lifelong learning and continuous personal and professional development. In today's rapidly evolving world, where new technologies and industries emerge at an unprecedented pace, individuals must be equipped with the necessary skills to adapt and thrive in a dynamic labor market. Tertiary education plays a crucial role in equipping individuals with the skills and knowledge needed to navigate these challenges and contribute to the growth and prosperity of their nation.

It is worth noting that the gross ratio of school enrollment in tertiary education is just one piece of the puzzle when it comes to assessing a country's knowledge and educational landscape. Other factors, such as the quality of education, the availability of resources, and the inclusivity of educational opportunities, also play a crucial role in shaping a country's overall educational profile. However, the gross enrollment ratio serves as a fundamental metric that provides a broad overview of a country's commitment to education and its efforts to develop a knowledgeable society.

In conclusion, the gross ratio of school enrollment in tertiary education serves as a valuable indicator of a country's knowledge and educational attainment. By analyzing this ratio, we can gain insights into the level of engagement in higher learning within a country's population. Moreover, a high gross enrollment ratio is associated with higher levels of innovation and reflects a commitment to lifelong learning and personal development. It is crucial for countries to prioritize and invest in tertiary education to foster a knowledgeable society that can drive innovation, contribute to societal progress, and adapt to the ever-changing demands of the global economy.

The final control variable is “the governance quality of a country” (**gov**). Jia et al. (2019) depicted that a country with high quality in public governance discourages the risk from innovation. This means innovation activities in a country have better conditions to develop with a good quality system of governance. Tee, Lain-Tze et al. (2014) also took “the governance quality of a country” into consideration to analyze the relationship between innovation and financial markets. Furthermore, Kaufmann et al. (2011) provided the components which are to calculate “gov” and the meaning behind its value, that is higher values, better governance quality. In particular, there are six elements of governance with a range scale of values from negative 2.5 to positive 2.5, as following: Control of corruption, Government effectiveness, Political stability and absence of violence or terrorism, Regulatory quality, Rule of law, Voice and Accountability.

- **Control of Corruption:** This element assesses the extent to which corruption is controlled within a government. It examines the presence of bribery, embezzlement, and other forms of corrupt practices. A higher value indicates a stronger commitment to combating corruption and promoting transparency.
- **Government Effectiveness:** This element measures the overall effectiveness of the government in implementing policies and delivering public services. It considers factors such as policy formulation, implementation capacity, and efficiency in service delivery. A higher value indicates a more efficient and capable government.
- **Political Stability and Absence of Violence or Terrorism:** This element evaluates the level of political stability and the absence of violence or terrorism within a country. It considers factors such as the likelihood of political unrest, the presence of armed conflicts, and the occurrence of terrorist activities. A higher value indicates a more stable and peaceful political environment.
- **Regulatory Quality:** This element focuses on the quality of regulations and their enforcement. It assesses the effectiveness of regulations in promoting market competition, protecting property rights, and ensuring the overall stability of the business environment. A higher value indicates better regulatory practices and enforcement.

- **Rule of Law:** This element measures the extent to which the government upholds the rule of law and ensures equal access to justice. It examines factors such as the independence of the judiciary, the protection of property rights, and the effectiveness of the legal system. A higher value indicates a stronger adherence to the rule of law.
- **Voice and Accountability:** This element assesses the ability of citizens to participate in the political process, express their views, and hold the government accountable. It considers factors such as freedom of speech, access to information, and the presence of a robust civil society. A higher value indicates a more inclusive and accountable governance system.

These six elements of governance provide a comprehensive framework for evaluating the overall quality of governance within a country. By understanding and addressing the challenges and opportunities associated with each element, governments can strive towards building stronger and more effective governance systems that serve the needs of their citizens.

Table 1. Summary of variables

Variables	Sources	Measurements	Expression in research model
Dependent variable			
The number of patent applications	World Development Indicators (WDI)	In logarithmic form	lpa
Banking sector development variables		In logarithmic form and is sum of ldeposit, lasset, lbank, lcredit	lbsd
Bank credit to bank deposits	Global Financial Development	In logarithmic form	ldeposit
Deposit money banks assets to GDP			lasset
Private credit by deposit money banks to GDP			lbank
Private credit by deposit money banks and other financial institutions to GDP			lcredit
Control variables			
GDP per capita	WDI	In logarithmic form	lgdp
Research and development expenditure	WDI	Expressed as a percent of GDP	rd
The gross ratio school enrollment in tertiary education	WDI	In logarithmic form	ledu
The governance quality of a country	Worldwide Governance Indicators	The average value of six governance elements	gov

Source: Own processing

3.3. Research model

Based on collected, processed data and the objective of identifying the effects of the development of banking system on innovative activities, this research structures the model as below:

$$lpa_{it} = \beta_0 + \beta_1 l gdp_{it} + \beta_2 rd_{it} + \beta_3 ledu_{it} + \beta_4 gov_{it} + \beta_5 lbsd_{it-1} + \alpha_i + \mu_{it}$$

where:

β : estimated coefficients

α : unobserved countries

μ : idiosyncratic disturbances

$i = 1, \dots, N$: countries

$t = 1, \dots, T$: years

In specific, this model mainly examines the link between banking sector development and patent applications number.

3.4. Research method and adjusted research model

This research uses a panel data analysis to find out how the development of banking system affects on innovation-related activities. Based on theories from Damodar (2009), Arellano et al. (1995) and Blundell et al. (1998), methods have been used to deal with the collected data include the pooled Ordinary Least Squares (OLS), the Fixed Effects Model (FEM), the Random Effects Model (REM) and the two-step system GMM.

The pooled OLS, also known as the classical linear regression model, assumes that the error term is uncorrelated with the explanatory variables and has a constant variance. This method is widely used when there is no concern about unobserved heterogeneity across individuals or time periods. However, it does not account for individual-specific effects or time-varying factors.

In contrast, the FEM is suitable when dealing with panel data that exhibit individual-specific effects. It allows for the estimation of individual-specific intercepts, capturing unobserved heterogeneity across individuals. This method effectively eliminates time-invariant unobserved individual-specific effects by including individual-specific dummy variables. However, it assumes that the individual-specific effects are uncorrelated with the explanatory variables.

The REM, on the other hand, treats the individual-specific effects as random variables following a specific distribution. This model assumes that the individual-specific effects are uncorrelated with the explanatory variables and have a constant variance. The REM is particularly useful when the individual-specific effects are correlated with the explanatory variables, as it provides consistent estimates of the parameters of interest.

Another approach commonly used in dynamic panel data models is the two-step system GMM. This method addresses endogeneity issues by employing instrumental variables and moment conditions. The first step involves estimating the model using a set of predetermined variables as instruments. In the second step, the resulting residuals are used to construct moment conditions, which are used to obtain efficient estimates of the parameters.

It is important to carefully consider the appropriate research method based on the specific characteristics of the data and the research objectives. Each method has its strengths and limitations, and the choice of method should be guided by theoretical considerations and empirical evidence.

Next, a summary of the process of analyzing the collected data is revealed as follows. The model is firstly estimated by the pooled Ordinary Least Squares (OLS). In case of taking country-specific attributes into consideration, the Fixed Effects Model (FEM) or Random Effects Model (REM) would be preferred depends on the results from the Hausman specification test. In addition, all classical linear model assumptions must be hold with any finally chosen method (OLS, FEM or REM). However, these methods fail to cope with the endogeneity problem in which a regressor variable is correlated with the residual term. Therefore, Arellano et al. (1995) and Blundell et al. (1998) proposed a method named Generalized Method of Moments (GMM) which not only figures out the endogenous variables problem, but also deal with the potential problem of heteroskedasticity and autocorrelation in the idiosyncratic disturbances term. In this regard, this paper applies the two-step system GMM method relied on forming a dynamic short panel data (smaller T and larger N) by employing a 1-year lag on the regressand (**lpa**) as a regressor variable. Following this, the adjusted research model becomes:

$$lpa_{it} = \beta_0 + \beta_1 lpa_{it-1} + \beta_2 lgdp_{it} + \beta_3 rd_{it} + \beta_4 ledu_{it} + \beta_5 gov_{it} + \beta_6 lbsd_{it-1} + \alpha_i + \mu_{it}$$

By applying the method of two-step system GMM, endogenous variables are imposed lags to be used as instruments with GMM style. The number of instruments may vary among various models because of the selection process of instruments. The number of groups are the one of researched countries. Mileva (2007) suggested a technical rule of thumb which is to consistently keep the number of instruments less than or equal to that of groups.

After estimating the model by using the two-step system GMM, this research employs the Hansen test which has the null hypothesis is that “Overall validity of the instruments used”, to verify the validity of the instrument variables chosen in the process of applying the two-step system GMM (Arellano et al., 1991). Furthermore, the Arellano and Bond test for $AR^7(2)$ with null hypothesis of “No existence of autocorrelation in the errors term” is also used in this paper to confirm the absence of serial autocorrelation in the idiosyncratic disturbances term of a dynamic panel data (Labra and Torrecillas, 2018).

⁷ Autoregressive

4. LITERATURE REVIEW

As mentioned in the "Introduction" section, this chapter aims to provide background knowledge related to innovation activities and the development of the banking sector. In addition, a hypothesis is formulated to explore the potential impact of bank system development on innovative activities.

However, it is important to first discuss the possible influences of a country's wealth, expenditure on research and development (R&D), knowledge, and governance quality on its innovative activities. This discussion will provide valuable context for understanding the need for control variables in the research model, as described in the "Methodology" section.

To begin with, it has been observed that countries with higher GDP per capita tend to have a greater incentive to engage in innovation activities (Barbosa et al., 2011). This suggests a positive relationship between a nation's income and its involvement in innovation-related endeavors.

Furthermore, research and development (R&D) efforts play a crucial role in fostering innovation (Ganotakis and Love, 2011). For instance, a study by Un et al. (2010) examined the relationship between R&D collaborations and product innovation. By analyzing data from nearly 800 manufacturing companies, the study found that collaborations with universities and suppliers have a positive impact on product innovation. Therefore, it can be anticipated that increased expenditure on R&D will stimulate activities related to innovation.

Moreover, the breadth and depth of knowledge have been found to be instrumental in achieving or developing innovation (Zhou and Li, 2012). Similarly, Jiménez-Jiménez and Sanz-Valle (2011) identified that organizational learning enhances innovation based on data collected from 451 Spanish firms. These findings imply that a nation's level of education plays a significant role in fostering its innovation activities.

Lastly, recent studies have shed light on the potential connection between a country's governance quality and its innovation activities. For example, Shapiro et al. (2015) examined the impact of corporate governance on innovation performance, focusing on privately owned small and medium-sized enterprises in China. The study revealed a strong positive effect of corporate governance on overall innovation activity, particularly in terms of patenting activity. Another study conducted by Amore and Bennedsen (2016) explored the influence of corporate governance on a specific type of innovation, namely green innovation. The findings highlighted that ineffective corporate governance can act as a significant constraint on the efficiency of environmental innovation. Taking a broader perspective on governance, Tee, Lain-Tze et al. (2014) determined that countries with higher quality governance provide a more favorable environment for innovation activities. In summary, understanding the potential impact of a country's wealth, R&D expenditure, knowledge, and governance quality on its innovation activities is crucial for comprehending the complexities of the research model. By considering these factors, we can gain valuable insights into the dynamics of innovation and its relationship with the development of the banking sector.

Next, theoretical foundations about the first researched object of this paper which is innovation activity is presented as follows. Kline et al. (2010) showed innovation with varied features, it could be a new product, a more effective process of production, a substitution of cheaper material, a reorganization in general or an enhancement in methods.

From a different perspective, Kahn (2018) provides a comprehensive perspective on innovation, viewing it as a multifaceted concept with various dimensions. According to Kahn, innovation can be understood as an outcome, a process, and a mindset. By exploring each of these dimensions in more depth, we can gain a deeper understanding of the complexities and nuances of innovation.

Firstly, let's delve into innovation as an outcome. This dimension focuses on the potential outputs that can be achieved through innovation. These outputs encompass different aspects of innovation, including product innovation, process innovation, marketing innovation, business model innovation, supply chain innovation, and organizational innovation.

Product innovation involves the development and improvement of products, and it can take various forms. These forms include cost reductions, product improvements, line extensions, exploring new markets, discovering new uses for existing products, introducing new category entries, and even creating entirely new products that have never been seen before.

Process innovation, on the other hand, concentrates on enhancing the efficiency and effectiveness of systems within an organization. This can involve optimizing production processes, streamlining delivery services, and improving organizational processes to achieve higher levels of productivity and performance.

Marketing innovation is centered around attracting and acquiring more customers and consumers. Organizations employ various strategies to achieve this, such as increasing promotional efforts, creating unique and compelling products, and exploring innovative marketing channels and techniques.

Business model innovation refers to making significant changes to the underlying business model. This can include redefining the industry model, exploring new revenue models, and innovating the overall enterprise model to drive growth and sustainable competitive advantage.

Supply chain innovation encompasses three key elements: the supply chain network structure, supply chain technology, and supply chain business processes. By innovating these areas, organizations can enhance their supply chain capabilities, optimize logistics, and improve overall operational efficiency.

Lastly, organizational innovation focuses on transforming the organizational structure, management forms, and working environments. This dimension recognizes the importance of adapting organizational practices to foster a culture of innovation, empowering employees to think creatively, and promoting collaboration and experimentation.

Moving on to innovation as a process, Kahn describes it as a three-step model: discover, develop, and deliver. The process begins with the discovery phase, where ideas and opportunities are identified and evaluated. These ideas are then developed through experimentation, prototyping, and refinement. Finally, the innovations are delivered to the market or implemented within the organization to generate value and impact.

In addition to understanding innovation as an outcome and a process, Kahn emphasizes the significance of innovation as a mindset. This dimension highlights the importance of individual members' awareness, attitudes, and behaviors towards innovation. By fostering a culture of innovation, organizations can create an environment where creativity, risk-taking, and continuous improvement are encouraged and rewarded.

In conclusion, innovation is a multifaceted concept that encompasses various dimensions. By recognizing innovation as an outcome, a process, and a mindset, organizations can develop a holistic approach to drive and sustain innovation. Embracing innovation in all its dimensions can lead to enhanced competitiveness, growth, and success in today's dynamic and rapidly evolving business landscape.

From these innovation-related studies above, it is obvious that innovation was possibly understood by different ways in vary contexts. Thus, an expectation for existence of various innovation measurements is reasonable.

According to Barbosa et al. (2011), they measured innovation under a viewpoint of innovation intensity at the industry level across European countries with institutional differences which are product market regulation, labor market regulation, capital market regulation and intellectual property rights. The interesting result of this study is that intellectual property protection affects negatively on the innovation intensity at the industry level across European countries, although the empirical researches expected that improving intellectual property rights encourages innovation. However, this research also found another outcome which is more valuable and generally consistent with previous literatures, the capital market development strengthens innovation. Moreover, the development of capital market is measured as a developed credit market.

Likewise, in a reliable research on a wide range of 44 countries conducted by Ang (2011), it generalized the developed credit market as the financial development by using indicator as the ratio of private credit to GDP. This explains there is a close link between a country's bank market development and financial market expansion.

Discussing specifically about the financial development, it is divided into bank market development and stock market development (Sadorsky, 2011; Tee et al., 2014). While, Adnan (2011) explained it in a more complex approach but with greater details also. The study expressed financial development as risk diversification and effective capital allocation. It is also observed by evaluating the performance of banking systems, stock markets, insurance companies and bond markets with evidence collected from over 40 economies in the period from 1988 to 2009.

Accordingly, discovering a nation's financial development in comprehension requires a greater effort and deep knowledge for the four mentioned markets. Nonetheless, with limited resources as mentioned in the "Methodology" part, this paper aims to examine only the development of banking system and its impacts on innovative activities. Therefore, next literatures focus on introducing theoretical framework related to banking sector and its development.

Fiordelisi et al. (2011) observed a set of data from European commercial banks and provided some valuable information about European banking efficiency. In particular, the concept of bank efficiency should be examined separately into cost and revenue efficiency rather than profit efficiency only. Moreover, cost efficiency directly leads to higher level of bank capital. This means more efficient banks eventually become better capitalized. Additionally, the study interestingly indicated the existence of reverse effect that higher bank capital levels also tend to positively impact on the level of bank efficiency.

Approaching to banking sector with similar concept with “bank efficiency”, Bose et al. (2012) estimated “banking development” based on credit provided to the private sector, total domestic credit provided by the banking sector and liquid liabilities, all expressed as percentage of GDP. Likewise, Aslan et al. (2014) measured banking development via four indicators, as percentage of GDP: deposit money bank assets, financial system deposits, liquid liabilities, private credit by deposit money banks and other financial institutions. From these studies, the method of evaluating banking development reveals an implication that the credit supply and the ability to access to it determine the development of bank system.

The following researches contributes the first direct visions to the association between credit supply and innovative activities. Amore et al. (2013) indicated significant advantages in quantity and quality of innovation activities are attained with better credit supply, the research scope is in the United States in approximately 10 years. Correspondingly but with larger and more specific data of over 19,000 firms across 47 developing countries, Ayyagari et al. (2011) found the similar relationship, accessing to the external financing brings better opportunities to foster firm innovation. Moreover, these firms are classified with greater details as the two groups that are public and private enterprises, and small and medium-sized corporates.

Other studies support the above association with another perspective of banking system development or credit supply, which are financial constraints. Gorodnichenko et al. (2013) implied domestic firms innovation is restrained by financial barriers. This result is based on large surveys of nearly 8,000 firms in 27 different countries. More specifically, García-Quevedo et al. (2018) analyzed it with data from Spanish firms and proved that financial obstacles increase the possibility of failing an innovation project. Using a different approach, Ding et al. (2022) considered fintech as a new dimension of financial development in recent years, then found that fintech has a positive impact on firm innovation and more significantly positive effects on firms with currently financial constraints.

Based on above discussions, in case of viewing the development of banking sector, generalizing it as financial development and observing its impacts on innovation from macro to micro, the following literatures support the idea that financial development stimulates innovation.

First, Paunov (2012) indicated that the global crisis results in increasing financial constraints, then many firms stop ongoing innovation projects. Apparently, a globally financial events indirectly influence on innovation activities.

Second, with smaller perspective of financial development, Maskus et al. (2012) examined the effects of national and international financial markets on R&D intensity and revealed that the effectiveness of domestic and foreign financial markets plays a crucial role in encouraging innovation in general and R&D in specific.

Last, Meierrieks (2014) researched financial development with the view of cross-country only, used a dataset of over 50 countries in the period from 1993 to 2008 and showed the similar association, a more effectively innovative activity corresponding to the higher levels of a nation's financial development.

These discussed studies above lead to possibility that the domestic banking sector development contributes to the development of financial market, subsequently encourages innovation activities. Therefore, the main hypothesis of this paper is established concisely as follows:

“Banking sector development is positively associated with innovation activities”

5. RESULTS

This chapter includes outcomes of econometric analysis, its detailed interpretation and discussion about the findings.

5.1. Econometrics results

The OLS regression is performed at first, followed by the verification of potential problems of models such as multicollinearity by using the value of mean VIF⁸, heteroskedasticity in errors term by executing the White's test and autocorrelation in residuals term by applying the Wooldridge test (Damodar, 2009; Drukker, 2003).

Subsequently, the two-step system GMM is employed to deal with the endogenous variables problem besides heteroskedasticity in the idiosyncratic disturbances term which is revealed from the results of White's test.

The estimated results exported from Stata software are listed as below.

⁸ Variance inflation factor

Model 1 is structured with independent variables are control ones only.

Table 2 depicts the OLS regression results for Model 1. It confirms that 41% of the model is explained by the independent variables.

Table 2. OLS regression for Model 1

Source	SS	df	MS	Number of obs	264
Model	404.87	4	101.22	F(4, 259)	44.72
Residual	586.19	259	2.26	Prob > F	0.00
				R-squared	0.41
				Adj R-squared	0.40
Total	991.06	263	3.77	Root MSE	1.50

lpa	Coef.	Std. Err.	t	P>t	[95% Conf. Interval]
lgdp	0.91	0.42	2.15	0.03**	0.08 1.75
rd	2.03	0.19	10.60	0.00***	1.65 2.41
ledu	0.01	0.36	0.02	0.98	-0.70 0.72
gov	-2.60	0.38	-6.92	0.00***	-3.34 -1.86
_cons	-3.71	5.05	-0.73	0.46	-13.66 6.24

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The table also shows that only the coefficients of “GDP per capita” and “Research and development expenditure” are significant with the expected positive directions.

Accordingly, multicollinearity, heteroskedasticity and autocorrelation are suspected to exist.

Table 3 gives the results of testing for multicollinearity for Model 1. Variance inflation factors is used and the criteria is that there is multicollinearity if the variance inflation factor is higher than 10.

Table 3. Statistical test for checking multicollinearity for Model 1

Variable	VIF	1/VIF
gov	4.23	0.24
rd	3.49	0.29
lgdp	3.11	0.32
ledu	1.71	0.58
Mean VIF	3.14	

The results above show that all variance inflation factors are less than 10. This indicates that the model 1 is free from multicollinearity.

Table 4 presents the White's test result for checking heteroskedasticity in errors term for Model 1. The criteria is that there is heteroskedasticity in errors term if the "Prob>chi2" is less than 5%.

Table 4. White's test for checking heteroskedasticity for Model 1

Source	chi2	df	Prob > chi2
Heteroskedasticity	81.22	14	0.00
Skewness	9.01	4	0.06
Kurtosis	0.45	1	0.50
Total	90.68	19	0.00

Obviously, the results show the "Prob>chi2" is less than 5%. This implies the model 1 is in the problem of heteroskedasticity in the errors term.

Table 5 displays the result of the Wooldrige test for checking autocorrelation for Model 1. The criteria is that there is autocorrelation if the "Prob > F" is less than 5%.

Table 5. Wooldrige test for checking autocorrelation for Model 1

F(1, 23)	0.17
Prob > F	0.69

As can be seen from the result, the "Prob>F" is greater than 5%. This concludes there is no existence of autocorrelation in the model's errors term.

After detection of heteroskedasticity in the idiosyncratic disturbances term, the two-step system GMM is applied to deal with it and potential endogenous variables problem. Thus, Table 6 shows the results from applying the two-step system GMM for Model 1.

Table 6. Two-step system GMM's result for Model 1

Group variable	country_code	Number of obs	240
Time variable	Year	Number of groups	24
Number of instruments	21	Obs per group: min	10
Wald chi2(4)	177975.51	avg	10
Prob > chi2	0.00	max	10

lpa	Coef.	Std. Err.	z	P>z	[95% Conf. Interval]
lpa L1.	0.99	0.00	278.39	0.00***	0.99 1.01
lgdp	0.14	0.02	6.28	0.00***	0.10 0.19
rd	-0.12	0.02	-5.71	0.00***	-0.17 -0.08
ledu	0.12	0.03	4.27	0.00***	0.07 0.18
gov	0.14	0.03	4.80	0.00***	0.08 0.20
_cons	-1.97	0.27	-7.23	0.00***	-2.51 -1.44

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The estimated coefficients state that three out of four control variables are positively related to “The number of patent applications”. Specifically, its interpretations as follows. 0.14 is the expected percentage increase of “The number of patent applications”, if “GDP per capita” rises by 1%. 0.12 is the expected percentage decrease of “The number of patent applications”, if “R&D expenditure, expressed as a percentage of GDP” rises by 1 unit. 0.12 is the expected percentage increase of “The number of patent applications”, if “The gross ratio school enrollment in tertiary education” rises by 1%. 0.14 is the expected percentage increase of “The number of patent applications”, if “The governance quality of a country” rises by 1 unit.

Following the two-step system GMM, table 7 gives the results of Arellano-Bond test for AR(2) and Hansen test for the model 1. These criteria are that there is autocorrelation if “Pr>z” is less than 5%, and the used instrument variables are valid if the “Prob>chi2” is greater than 5%

Table 7. Arellano-Bond test for AR(2) and Hansen test for the model 1

Arellano-Bond test for AR(2) in first differences	$z = 0.94$	$\text{Pr} > z = 0.35$
Hansen test of overid. restrictions	$\text{chi2}(15) = 16.93$	$\text{Prob} > \text{chi2} = 0.32$

The result for the Arellano-Bond test for AR(2) shows that the “Pr>z” is greater than 5%. This verifies that the model is not in the problem of serial autocorrelation in the residuals term.

The result for the Hansen test shows that the “Prob>chi2” is greater than 5%. This implies that the model is fixed by the two-step system GMM with valid instrument variables.

Model 2 is designed from Model 1 with an additionally independent variable representing for banking sector development that is “Private credit by deposit money banks to GDP” employed 1-year lag.

Table 8 depicts the OLS regression results for Model 2. It confirms that 46% of the model is explained by the independent variables.

Table 8. OLS regression for Model 2

Source	SS	df	MS			
Model	460.35	5	92.07	Number of obs	264	
Residual	530.72	258	2.06	F(5, 258)	44.76	
Total	991.06	263	3.77	Prob > F	0.00	
				R-squared	0.46	
				Adj R-squared	0.45	
				Root MSE	1.43	

lpa	Coef.	Std. Err.	t	P>t	[95% Conf. Interval]
lgdp	0.69	0.41	1.70	0.09*	-0.11 1.49
rd	2.18	0.18	11.81	0.00***	1.82 2.55
ledu	-0.47	0.36	-1.32	0.19	-1.17 0.23
gov	-3.07	0.37	-8.32	0.00***	-3.80 -2.34
lbank_t_1	1.23	0.24	5.19	0.00***	0.76 1.69
_cons	-4.46	4.82	-0.93	0.36	-13.96 5.03

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The table also shows that the coefficients of “The gross ratio school enrollment in tertiary education” is insignificant and “The governance quality of a country” is significant but with the unexpected direction. Therefore, the model is considered to meet the problem of multicollinearity, heteroskedasticity and autocorrelation.

Table 9 gives the results of testing for multicollinearity for Model 2. Variance inflation factors is used and the criteria is that there is multicollinearity if the variance inflation factor is higher than 10.

Table 9. Statistical test for checking multicollinearity for Model 2

Variable	VIF	1/VIF
gov	4.50	0.22
rd	3.59	0.28
lgdp	3.14	0.32
ledu	1.84	0.54
lbank_t_1	1.22	0.82
Mean VIF	2.86	

The results above show that all variance inflation factors are less than 10. This result points out that the model is not having multicollinearity problem.

Table 10 presents the White's test result for checking heteroskedasticity in errors term for Model 2. The criteria is that there is heteroskedasticity in errors term if the “Prob>chi2” is less than 5%.

Table 10. White's test for checking heteroskedasticity for Model 2

Source	chi2	df	Prob > chi2
Heteroskedasticity	108.89	20	0.00
Skewness	9.00	5	0.11
Kurtosis	0.09	1	0.76
Total	117.99	26	0.00

Obviously, the results show the “Prob>chi2” is less than 5%. This depicts that there is existence of heteroskedasticity in the idiosyncratic disturbances term of the model.

Table 11 displays the result of the Wooldrige test for checking autocorrelation for Model 2. The criteria is that there is autocorrelation if the “Prob > F” is less than 5%.

Table 11. Wooldrige test for checking autocorrelation for Model 2

F(1, 23)	0.21
Prob > F	0.65

As can be seen from the result, the “Prob>F” is greater than 5%. This concludes there is no existence of autocorrelation in the model’s errors term.

After the reveal of heteroskedasticity in the idiosyncratic disturbances term, the two-step system GMM is applied to deal with it and potential problem of endogenous variables. Thus, Table 12 shows the results from applying the two-step system GMM for Model 2.

Table 12. Two-step system GMM's result for Model 2

Group variable	country_code	Number of obs	240
Time variable	Year	Number of groups	24
Number of instruments	23	Obs per group: min	10
Wald chi2(5)	49417.47	avg	10
Prob > chi2	0.00	max	10

lpa	Coef.	Std. Err.	z	P>z	[95% Conf. Interval]
lpa L1.	0.97	0.02	58.09	0.00***	0.94 1.00
lgdp	0.34	0.09	3.81	0.00***	0.16 0.51
rd	-0.09	0.08	-1.17	0.24	-0.24 0.06
ledu	0.13	0.07	1.78	0.08*	-0.01 0.26
gov	-0.04	0.15	-0.25	0.81	-0.32 0.25
lbank_t_1	0.54	0.14	3.95	0.00***	0.27 0.80
_cons	-6.06	1.23	-4.94	0.00***	-8.47 -3.65

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The main purpose of this model 2 is to find out the influence of “Private credit by deposit money banks to GDP” on “The number of patent applications”. Thus its coefficient is interpreted as follows. 0.54 is the expected percentage increase of “The number of patent applications”, if “Private credit by deposit money banks to GDP” rises by 1%.

Following the two-step system GMM, table 13 gives the results of Arellano-Bond test for AR(2) and Hansen test for the model 2. These criteria are that there is autocorrelation if “Pr>z” is less than 5%, and the used instrument variables are valid if the “Prob>chi2” is greater than 5%

Table 13. Arellano-Bond test for AR(2) and Hansen test for the model 2

Arellano-Bond test for AR(2) in first differences	$z = 0.94$	$\text{Pr} > z = 0.35$
Hansen test of overid. restrictions	$\text{chi2}(15) = 20.34$	$\text{Prob} > \text{chi2} = 0.21$

The result for the Arellano-Bond test for AR(2) shows that the “Pr>z” is greater than 5%. This determines that the model is free from the problem of autocorrelation in the errors term.

The result for the Hansen test shows that the “Prob>chi2” is greater than 5%. This indicates that the model using the method of two-step system GMM with valid instrument variables.

Model 3 is continually established from Model 1 with another additionally independent variable standing for the growth of banking sector, that is “Deposit money banks assets to GDP” imposed 1-year lag.

Table 14 depicts the OLS regression results for Model 3. It confirms that 48% of the model is explained by the independent variables.

Table 14. OLS regression for Model 3

Source	SS	df	MS	Number of obs	264
Model	478.02	5	95.60	F(5, 258)	48.08
Residual	513.05	258	1.99	Prob > F	0.00
				R-squared	0.48
				Adj R-squared	0.47
Total	991.06	263	3.77	Root MSE	1.41

lpa	Coef.	Std. Err.	t	P>t	[95% Conf. Interval]
lgdp	0.60	0.40	1.48	0.14	-0.19 1.39
rd	2.10	0.18	11.68	0.00***	1.75 2.46
ledu	-0.40	0.34	-1.17	0.24	-1.08 0.28
gov	-2.86	0.35	-8.06	0.00***	-3.55 -2.16
lasset_t_1	1.51	0.25	6.06	0.00***	1.02 2.00
_cons	-5.33	4.75	-1.12	0.26	-14.67 4.02

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The table also shows that only the coefficients of “Research and development expenditure” and “Deposit money banks assets to GDP” are significant with the anticipated directions. Accordingly, testing for multicollinearity, heteroskedasticity and autocorrelation are necessary to be applied.

Table 15 gives the results of testing for multicollinearity for Model 3. Variance inflation factors is used and the criteria is that there is multicollinearity if the variance inflation factor is higher than 10.

Table 15. Statistical test for checking multicollinearity for Model 3

Variable	VIF	1/VIF
gov	4.29	0.23
rd	3.51	0.28
lgdp	3.16	0.32
ledu	1.78	0.56
lasset_t_1	1.13	0.88
Mean VIF	2.78	

The results above show that all variance inflation factors are less than 10. This argues that no multicollinearity problem appears in the model.

Table 16 presents the White's test result for checking heteroskedasticity in errors term for Model 3. The criteria is that there is heteroskedasticity in errors term if the “Prob>chi2” is less than 5%.

Table 16. White's test for checking heteroskedasticity for Model 3

Source	chi2	df	Prob > chi2
Heteroskedasticity	114.12	20	0.00
Skewness	17.02	5	0.00
Kurtosis	0.00	1	0.98
Total	131.15	26	0.00

Obviously, the results show the “Prob>chi2” is less than 5%. This implies there is no heteroskedasticity in the idiosyncratic disturbances term.

Table 17 displays the result of the Wooldrige test for checking autocorrelation for Model 3. The criteria is that there is autocorrelation if the “Prob > F” is less than 5%.

Table 17. Wooldrige test for checking autocorrelation for Model 3

F(1, 23)	0.21
Prob > F	0.65

As can be seen from the result, the “Prob>F” is greater than 5%. This concludes that the model’s errors terms are not in the problem of autocorrelation.

After finding of heteroskedasticity in the idiosyncratic disturbances term, the two-step system GMM is applied to cope with it and potential endogenous variables problem.

Thus, Table 18 shows the results from applying the two-step system GMM for Model 3.

Table 18. Two-step system GMM's result for Model 3

Group variable	country_code	Number of obs	240
Time variable	Year	Number of groups	24
Number of instruments	22	Obs per group: min	10
Wald chi2(5)	15361.11	avg	10
Prob > chi2	0.00	max	10

lpa	Coef.	Std. Err.	z	P>z	[95% Conf. Interval]
lpa L1.	0.92	0.03	35.15	0.00***	0.87 0.97
lgdp	0.34	0.11	3.26	0.00***	0.14 0.55
rd	-0.03	0.08	-0.42	0.67	-0.20 0.13
ledu	0.64	0.27	2.34	0.02**	0.10 1.17
gov	-0.11	0.16	-0.69	0.49	-0.43 0.21
lasset_t_1	0.98	0.26	3.83	0.00***	0.48 1.49
_cons	-10.04	2.94	-3.41	0.00***	-15.81 -4.27

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

This model 3 mainly aims to examine the influence of “Deposit money banks assets to GDP” on “The number of patent applications”. Therefore, the interpretation of its coefficient is as follows. 0.98 is the expected percentage increase of “The number of patent applications”, if “Deposit money banks assets to GDP” rises by 1%.

Following the two-step system GMM, table 19 gives the results of Arellano-Bond test for AR(2) and Hansen test for the model 3. These criteria are that there is autocorrelation if “Pr>z” is less than 5%, and the used instrument variables are valid if the “Prob>chi2” is greater than 5%

Table 19. Arellano-Bond test for AR(2) and Hansen test for the model 3

Arellano-Bond test for AR(2) in first differences	$z = 0.92$	$\text{Pr} > z = 0.36$
Hansen test of overid. restrictions	$\text{chi2}(15) = 14.66$	$\text{Prob} > \text{chi2} = 0.48$

The result for the Arellano-Bond test for AR(2) shows that the “Pr>z” is greater than 5%. This proves that the model is not having the problem of autocorrelation in the errors term. The result for the Hansen test shows that the “Prob>chi2” is greater than 5%. This demonstrates that the two-step system GMM is applied with valid instrument variables.

Model 4 is also set from Model 1 with an added independent variable which is “Bank credit to bank deposits” applied a year lag, representing for the development of banking system.

Table 20 depicts the OLS regression results for Model 4. It confirms that 47% of the model is explained by the independent variables.

Table 20. OLS regression for Model 4

Source	SS	df	MS	Number of obs	264
Model	466.58	5	93.32	F(5, 258)	45.90
Residual	524.48	258	2.03	Prob > F	0.00
				R-squared	0.47
				Adj R-squared	0.46
Total	991.06	263	3.77	Root MSE	1.43

lpa	Coef.	Std. Err.	t	P>t	[95% Conf. Interval]
lgdp	2.41	0.49	4.96	0.00***	1.45 3.37
rd	1.89	0.18	10.34	0.00***	1.53 2.26
ledu	-0.85	0.38	-2.28	0.02**	-1.59 -0.12
gov	-3.20	0.37	-8.59	0.00***	-3.93 -2.46
ldeposit_t_1	1.82	0.33	5.51	0.00***	1.17 2.48
_cons	-23.41	5.98	-3.92	0.00***	-35.18 -11.64

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The table also shows that the coefficients of “The gross ratio school enrollment in tertiary education” and “The governance quality of a country” are significant but with the unexpected movements. Therefore, multicollinearity, heteroskedasticity and autocorrelation are suspected to appear.

Table 21 gives the results of testing for multicollinearity for Model 4. Variance inflation factors is used and the criteria is that there is multicollinearity if the variance inflation factor is higher than 10.

Table 21. Statistical test for checking multicollinearity for Model 4

Variable	VIF	1/VIF
gov	4.62	0.22
lgdp	4.53	0.22
rd	3.56	0.28
ldeposit_t_1	2.44	0.41
ledu	2.07	0.48
Mean VIF	3.44	

The results above show that all variance inflation factors are less than 10. This considers the model is not in the problem of multicollinearity.

Table 22 presents the White's test result for checking heteroskedasticity in errors term for Model 4. The criteria is that there is heteroskedasticity in errors term if the “Prob>chi2” is less than 5%.

Table 22. White's test for checking heteroskedasticity for Model 4

Source	chi2	df	Prob > chi2
Heteroskedasticity	83.23	20	0.00
Skewness	13.94	5	0.02
Kurtosis	2.80	1	0.09
Total	99.97	26	0.00

Obviously, the results show the “Prob>chi2” is less than 5%. This indicates heteroskedasticity in the idiosyncratic disturbances term is found in the model.

Table 23 displays the result of the Wooldrige test for checking autocorrelation for Model 4. The criteria is that there is autocorrelation if the “Prob > F” is less than 5%.

Table 23. Wooldrige test for checking autocorrelation for Model 4

F(1, 23)	0.11
Prob > F	0.75

As can be seen from the result, the “Prob>F” is greater than 5%. This states that autocorrelation in the model’s errors term is not detected.

After the reveal of heteroskedasticity in the idiosyncratic disturbances term, the two-step system GMM is executed to fix it and potential problem of endogenous variables. Thus, Table 24 shows the results from applying the two-step system GMM for Model 4.

Table 24. Two-step system GMM's result for Model 4

Group variable	country_code	Number of obs	240
Time variable	Year	Number of groups	24
Number of instruments	22	Obs per group: min	10
Wald chi2(5)	492151.47	avg	10
Prob > chi2	0.00	max	10

lpa	Coef.	Std. Err.	z	P>z	[95% Conf.	Interval]
lpa L1.	0.99	0.00	245.21	0.00***	0.98	1.00
lgdp	0.25	0.04	6.81	0.00***	0.18	0.32
rd	-0.13	0.02	-5.19	0.00***	-0.18	-0.08
ledu	0.08	0.03	2.41	0.02**	0.01	0.14
gov	0.09	0.04	2.44	0.02**	0.02	0.17
ldeposit_t_1	0.12	0.03	3.80	0.00***	0.06	0.18
_cons	-3.31	0.44	-7.52	0.00***	-4.18	-2.45

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

This model 4 also mainly aims to find out the influence of “Bank credit to bank deposits” on “The number of patent applications”. Therefore, its coefficient is explained as follows. 0.12 is the expected percentage increase of “The number of patent applications”, if “Bank credit to bank deposits” rises by 1%.

Following the two-step system GMM, table 25 gives the results of Arellano-Bond test for AR(2) and Hansen test for the model 4. These criteria are that there is autocorrelation if “Pr>z” is less than 5%, and the used instrument variables are valid if the “Prob>chi2” is greater than 5%

Table 25. Arellano-Bond test for AR(2) and Hansen test for the model 4

Arellano-Bond test for AR(2) in first differences	$z = 0.94$	$\text{Pr} > z = 0.35$
Hansen test of overid. restrictions	$\text{chi2}(15) = 17.77$	$\text{Prob} > \text{chi2} = 0.28$

The result for the Arellano-Bond test for AR(2) shows that the “Pr>z” is greater than 5%. This confirms that no serial autocorrelation in the residuals term of the model.

The result for the Hansen test shows that the “Prob>chi2” is greater than 5%. This verifies that the model with the two-step system GMM is using valid instrument variables.

Model 5 is once again structured from Model 1 with an additionally independent variable standing for banking sector development that is “Private credit by deposit money banks and other financial institutions to GDP” imposed 1-year lag.

Table 26 depicts the OLS regression results for Model 5. It confirms that 47% of the model is explained by the independent variables.

Table 26. OLS regression for Model 5

Source	SS	df	MS	Number of obs	264
Model	468.55	5	93.71	F(5, 258)	46.27
Residual	522.52	258	2.03	Prob > F	0.00
				R-squared	0.47
				Adj R-squared	0.46
Total	991.06	263	3.77	Root MSE	1.42

lpa	Coef.	Std. Err.	t	P>t	[95% Conf. Interval]
lgdp	0.68	0.40	1.68	0.09*	-0.12 1.47
rd	2.19	0.18	11.94	0.00***	1.83 2.55
ledu	-0.48	0.35	-1.35	0.18	-1.17 0.22
gov	-3.08	0.37	-8.42	0.00***	-3.79 -2.36
lcredit_t_1	1.35	0.24	5.61	0.00***	0.88 1.82
_cons	-4.86	4.79	-1.02	0.31	-14.28 4.57

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

The table also shows that the coefficients of “The gross ratio school enrollment in tertiary education” is insignificant and “The governance quality of a country” is significant but with the unexpected movements. Thus, statistical testing for multicollinearity, heteroskedasticity and autocorrelation are essential to examine as follows.

Table 27 gives the results of testing for multicollinearity for Model 5. Variance inflation factors is used and the criteria is that there is multicollinearity if the variance inflation factor is higher than 10.

Table 27. Statistical test for checking multicollinearity for Model 5

Variable	VIF	1/VIF
gov	4.47	0.22
rd	3.58	0.28
lgdp	3.14	0.32
ledu	1.82	0.55
lcredit_t_1	1.20	0.83
Mean VIF	2.84	

The results above show that all variance inflation factors are less than 10. This implies the model is not caught at having multicollinearity.

Table 28 presents the White's test result for checking heteroskedasticity in errors term for Model 5. The criteria is that there is heteroskedasticity in errors term if the “Prob>chi2” is less than 5%.

Table 28. White's test for checking heteroskedasticity for Model 5

Source	chi2	df	Prob > chi2
Heteroskedasticity	117.47	20	0.00
Skewness	9.94	5	0.08
Kurtosis	0.03	1	0.87
Total	127.43	26	0.00

Obviously, the results show the “Prob>chi2” is less than 5%. This concludes the model is in the problem of heteroskedasticity in the errors term.

Table 29 displays the result of the Wooldrige test for checking autocorrelation for Model 5. The criteria is that there is autocorrelation if the “Prob > F” is less than 5%.

Table 29. Wooldrige test for checking autocorrelation for Model 5

F(1, 23)	0.22
Prob > F	0.65

As can be seen from the result, the “Prob>F” is greater than 5%. This confirms there is no existence of serial autocorrelation in the errors term of the model.

After finding of heteroskedasticity in the residuals term, the two-step system GMM is applied to face with it and problem of potential endogenous variables. Thus, Table 30 shows the results from applying the two-step system GMM for Model 5.

Table 30. Two-step system GMM's result for Model 5

Group variable	country_code	Number of obs	240
Time variable	Year	Number of groups	24
Number of instruments	23	Obs per group: min	10
Wald chi2(5)	54428.05	avg	10
Prob > chi2	0.00	max	10

lpa	Coef.	Std. Err.	z	P>z	[95% Conf. Interval]
lpa L1.	0.97	0.02	64.33	0.00***	0.94 1.00
lgdp	0.33	0.09	3.86	0.00***	0.16 0.50
rd	-0.10	0.07	-1.47	0.14	-0.24 0.03
ledu	0.14	0.07	2.09	0.04**	0.01 0.27
gov	-0.00	0.14	-0.01	0.99	-0.27 0.26
lcredit_t_1	0.51	0.12	4.08	0.00***	0.26 0.75
_cons	-5.94	1.19	-5.00	0.00***	-8.27 -3.61

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

This model 5 primarily examines the impact of “Private credit by deposit money banks and other financial institutions to GDP” on “The number of patent applications”. Therefore, its coefficient is explained as follows. 0.51 is the expected percentage increase of “The number of patent applications”, if “Private credit by deposit money banks and other financial institutions to GDP” rises by 1%.

Following the two-step system GMM, table 31 gives the results of Arellano-Bond test for AR(2) and Hansen test for the model 5. These criteria are that there is autocorrelation if “Pr>z” is less than 5%, and the used instrument variables are valid if the “Prob>chi2” is greater than 5%

Table 31. Arellano-Bond test for AR(2) and Hansen test for the model 5

Arellano-Bond test for AR(2) in first differences	$z = 0.94$	$\text{Pr} > z = 0.35$
Hansen test of overid. restrictions	$\text{chi2}(15) = 20.39$	$\text{Prob} > \text{chi2} = 0.20$

The result for the Arellano-Bond test for AR(2) shows that the “Pr>z” is greater than 5%. This concludes that the model is not having the problem of serial autocorrelation in the errors term.

The result for the Hansen test shows that the “Prob>chi2” is greater than 5%. This shows that the two-step system GMM is applied in the model with valid instrument variables.

5.2. Discussion

This part provides some arguments behind the econometric results and comparisons with other reliable researches. However, it is necessary to summarize all econometrics results in a table as below:

Table 32. Summary of econometric results

	Model 1	Model 2	Model 3	Model 4	Model 5
Mean VIF	3.14	2.86	2.78	3.44	2.84
White's test	0.00	0.00	0.00	0.00	0.00
(Prob > chi2)	[81.22]	[108.89]	[114.12]	[83.23]	[117.47]
Wooldridge test	0.69	0.65	0.65	0.75	0.65
(Prob > F)	0.17!	0.21!	0.21!	0.11!	0.22!
constant	-1.97	-6.06	-10.04	-3.31	-5.94
	(-7.23)	(-4.94)	(-3.41)	(-7.52)	(-5.00)
lgdp	0.14***	0.34***	0.34***	0.25***	0.33***
	(6.28)	(3.81)	(3.26)	(6.81)	(3.86)
rd	-0.12***	-0.09	-0.03	-0.13***	-0.10
	(-5.71)	(-1.17)	(-0.42)	(-5.19)	(-1.47)
ledu	0.12***	0.13*	0.64**	0.08**	0.14**
	(4.27)	(1.78)	(2.34)	(2.41)	(2.09)
gov	0.14***	-0.04	-0.11	0.09**	-0.00
	(4.80)	(-0.25)	(-0.69)	(2.44)	(-0.01)
lbank_t_1		0.54***			
		(3.95)			
lasset_t_1			0.98***		
			(3.83)		
ldeposit_t_1				0.12***	
				(3.80)	
lcredit_t_1					0.51***
					(4.08)
Number of instruments	21	23	22	22	23
Number of groups	24	24	24	24	24
Arellano and Bond test	0.35	0.35	0.36	0.35	0.35
for AR(2)	(0.94)	(0.94)	(0.92)	(0.94)	(0.94)
(Pr > z)					
Hansen test	0.32	0.21	0.48	0.28	0.20
(Prob > chi2)	[16.93]	[20.34]	[14.66]	[17.77]	[20.39]

Source: Own processing

***, ** and * depict significance at the level of 1%, 5% and 10% respectively

z-statistics revealed in parentheses, chi-square statistics indicated in square brackets, F-statistics followed by an exclamation mark.

In the section titled "Econometrics results" and according to Damodar (2009) and Drukker (2003), the regression models have been thoroughly analyzed to ensure the reliability of results. Firstly, it is worth noting that all the models are free from multicollinearity, as indicated by the VIF (Variance Inflation Factor) values of the independent variables, which are all less than 10. This ensures that there is no excessive correlation among the variables, and thus, the models can provide accurate estimations. Furthermore, the models exhibit no autocorrelation, as evidenced by the "Prob>F" values being greater than 5%. This implies that the error terms in the models are not dependent on each other over time. However, a challenge arises in the form of heteroskedasticity in the idiosyncratic disturbances term. The "Prob>Chi-square" values are found to be less than 5%, indicating the presence of heteroskedasticity. To address this issue, the author employs the two-step system GMM (Generalized Method of Moments) method in all the models, as explained in the preceding section.

Moving on to Model 1, this model focuses solely on control variables as independent variables. The objective here is to determine the impact of these control variables on the number of patent applications before introducing variables related to banking sector development into the regression model. The findings from Model 1 reveal that all the control variables have a significant influence on the number of patent applications. Particularly noteworthy is the positive relationship between a country's wealth, knowledge, and governance quality, and its innovative activities. These results are not isolated, as they are supported by previous studies conducted by Teitel (1994), Moon et al. (2019), and Jia et al. (2019). However, an intriguing discovery emerges in this paper, as it uncovers a negative impact of R&D expenditure on the number of patent applications. This suggests that a nation's R&D program may actually hinder its activities in the realm of innovation. Interestingly, Yoo and Park (2022) present a comprehensive converse, highlighting the significant contribution of a national innovation system through its R&D program. These contrasting perspectives add depth to the research and shed light on the complexities of the relationship between R&D expenditure and innovation.

The subsequent models, namely Model 2 through Model 5, focus on the impact of banking sector development on innovation activities. In each of these models, a specific variable related to the development of the banking system is individually added to Model 1. The coefficients of these variables, namely "Private credit by deposit money banks to GDP", "Deposit money banks assets to GDP", "Bank credit to bank deposits", and "Private credit by deposit money banks and other financial institutions to GDP" exhibit positive values and are highly significant. These findings align with previous studies conducted by Beck et al. (2010), Amore et al. (2013), and Tee, Lain-Tze et al. (2014), which have also demonstrated the positive relationship between the development of a nation's bank system and favorable conditions for innovation activities. This implies that a well-developed banking system plays a crucial role in fostering innovation. Moreover, it is noteworthy that the development of the banking system can be generalized as financial development (Ang, 2011). Therefore, the results of this research contribute to the implication that improvements in financial markets also encourage innovation-related activities, as highlighted in the studies by Meierrieks (2014), García-Quevedo et al. (2018), and Ding et al. (2022).

Furthermore, the general results regarding the impact of "GDP per capita" "Research and development expenditure" "The gross ratio school enrollment in tertiary education" and "The governance quality of a country" on the number of patent applications in Model 2 through Model 5 are consistent with the outcomes of Model 1. This consistency reinforces the notion that these control variables consistently play significant roles in influencing innovation activities.

Lastly, it is crucial to mention the validity of the models and their results. The table highlights that all the models adhere to the rule of maintaining the number of instruments equal to or less than the number of groups, as established by Mileva (2007). This ensures the validity of the models and contributes to the reliability of the findings. Additionally, the Arellano and Bond test for AR(2) with insignificant "Pr>z" further confirms the absence of serial autocorrelation in the idiosyncratic disturbances term of the dynamic panel data, as observed in the research by Labra and Torrecillas (2018). Furthermore, the Hansen test with insignificant "Prob > Chi-square" verifies the validity of the instrument variables, as demonstrated by Arellano et al. (1991).

6. CONCLUSION

In this research study, the main objective is to thoroughly examine the impact of cross-country banking sector development on the innovation activities within the selected EU countries. The study period spans 11 years, from 2010 to 2020, and includes data from 24 chosen EU countries.

The collected data is considered secondary and is treated as dynamic short panel data in the regression model. The regression model used in this study is a linear model and is solved using the two-step system GMM, which is a widely recognized and important method in the field.

The outcomes of the analysis reveal that a developed banking system in an EU country plays a significant role in supporting its innovative activities. This implies that a well-functioning banking sector can provide the necessary resources and financial support to foster innovation within the country. Moreover, the research also suggests that the development of financial markets within a country can have positive effects on its innovation activities. This underlines the importance of a comprehensive and integrated financial system in promoting innovation.

Based on the findings, the author proposes specific recommendations for the EU banking system. These recommendations prioritize certain ratios that have shown to have a positive impact on the number of patent applications. The ratios, ranked from highest to lowest level of positive impact, include Deposit money banks assets to GDP, Private credit by deposit money banks to GDP, Private credit by deposit money banks and other financial institutions to GDP, and Bank credit to bank deposits. By focusing on increasing these ratios, the EU banking system can create an environment that is more conducive to the development of innovative activities within EU countries.

It is important to note that the author acknowledges the limitations of the research. Specifically, the paper highlights certain data shortcomings that may have influenced the results. These limitations are discussed in detail in the "Methodology" section of the research. In addition, the author suggests potential areas for future research. One such area is the separation of a country's financial development into bank-based development and stock market development. Another interesting avenue for future studies is the consideration of up to four different markets, including banking systems, stock markets, insurance companies, and bond markets. Such investigations would provide further insights into the impact of each market on a nation's innovative activities, leading to more valuable and comprehensive proposals.

In conclusion, this research study sheds light on the crucial role played by the banking sector and financial markets in supporting innovation activities within EU countries. The findings and recommendations presented in this study provide valuable insights for policymakers and stakeholders in the banking and financial sectors, as well as for researchers and scholars interested in the relationship between banking sector development and innovation.

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